

EXPERIMENT - 1

TO SEPARATE THE INSOLUBLE COMPONENT IN PURE AND DRY STATE FROM THE GIVEN MIXTURE OF SOLUBLE AND INSOLUBLE SOLIDS (SODIUM CHLORIDE, SAND AND CAMPHOR)

APPARATUS REQUIRED

- | | |
|-----------------|--------------------|
| a) Beaker | b) Porcelain basin |
| c) Funnel | d) Filter Paper |
| e) Tripod stand | f) Burner |
| g) Sand Bath | h) Glass rod |
| i) Test tube | j) Spatula |

CHEMICAL REQUIRED

- Silver Nitrate Solution
- Mixture of sand, Common salt and Camphor

THEORY

Soluble Component - Those Component which completely dissolve in water is called Soluble Component. e.g. Salt (NaCl)

Non Soluble Component - Those Component which doesn't dissolve in water is called non-soluble component. e.g. Sand

Volatile Substances - Those substances that directly convert into vapour at room temperature is called volatile substances. e.g. Camphor

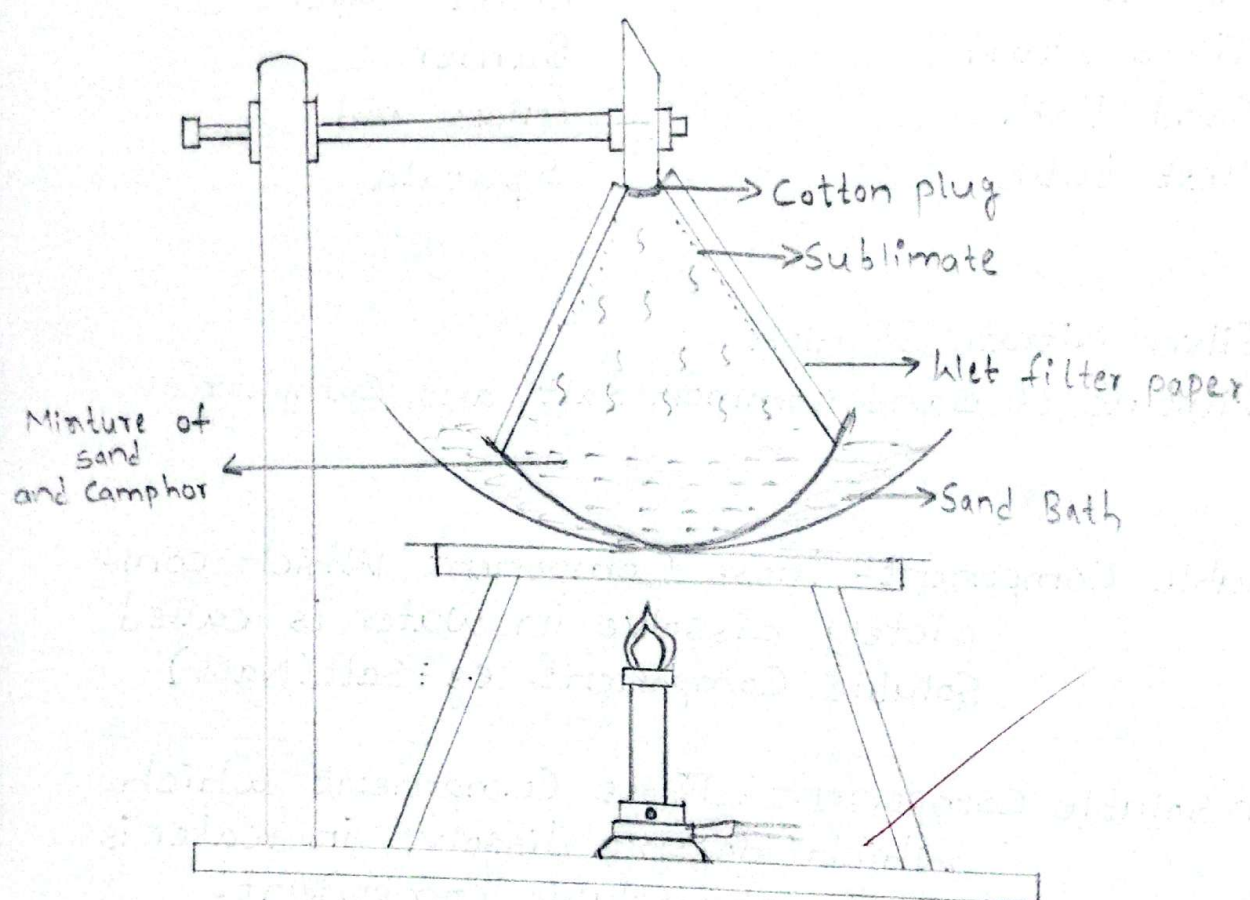


fig - Sublimation

Non Volatile substance - Those substances that do not directly convert into vapour at room temperature is called Non volatile substances. e.g: NaCl, Sand

Sublimation - The process of converting a volatile solid directly into vapour state by heating and again the vapour is converted back into solid state by cooling without passing into liquid state.

Sublimate - The condensed vapour that is obtained in the solid form during sublimation.

Filtration - The process of separating insoluble substance from soluble substance with the help of filter ~~ate~~ paper is called filtration.

Filtrate - The soluble substance which passes through the filter paper during filtration is called filtrate.

Residue - The insoluble substance that remains in the filter paper is known as residue.

PROCEDURE

The given mixture of sand, common salt and camphor was taken in a clean porcelain basin, in order to / for the separation of camphor.

A funnel was fixed over the porcelain basin and the mixture was heated. Camphor sublimed and was collected on the inner side of the funnel. After the complete sublimation of camphor,

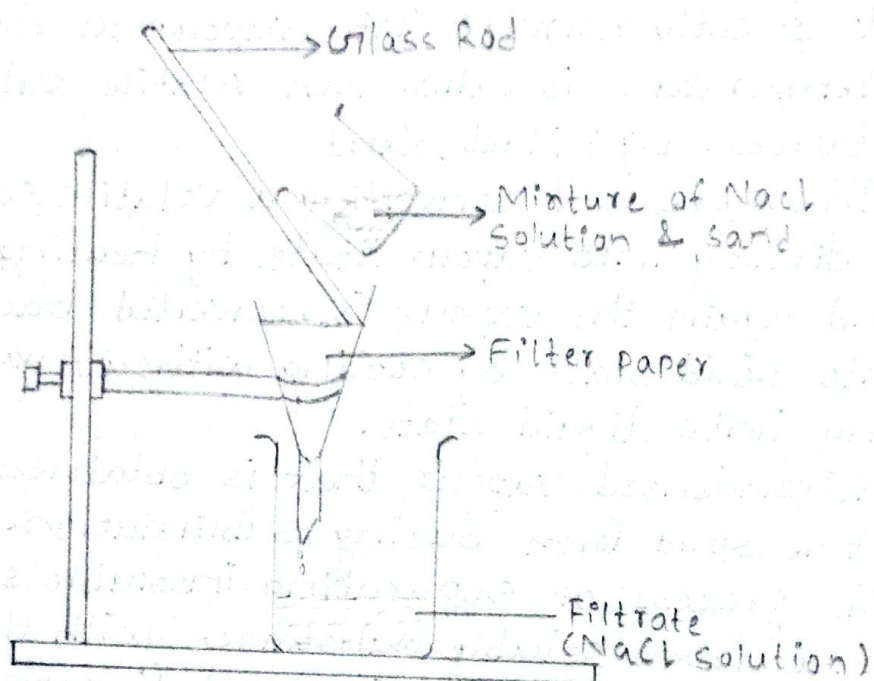


fig- filtration

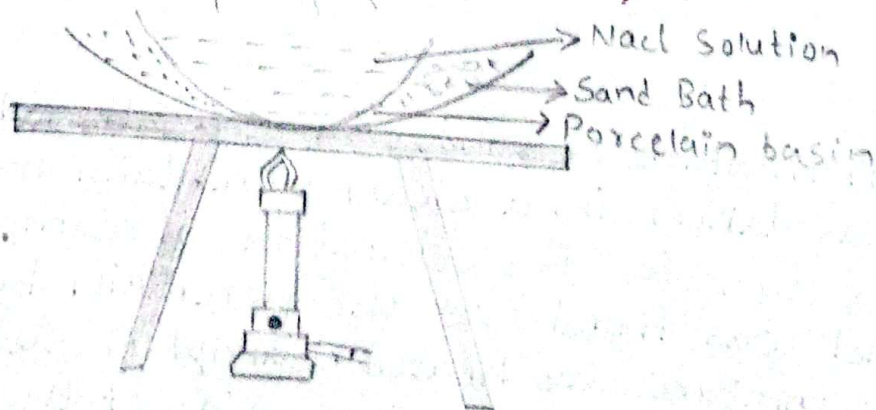


fig- Evaporation of NaCl Solution

the remaining mixture of sand and common salt was transferred into a beaker and sufficient water was added. The contents were stirred well with a glass rod so that the common salt dissolved properly. The mixture was subsequently filtered, leaving sand as residue on the filter paper. The residue (sand) was thoroughly washed repeatedly with water to remove any traces of common salt, and thus pure sand was obtained. The filtrate containing common salt solution was collected in a porcelain basin and was carefully heated to evaporate the water completely. In this way, pure crystals of common salt were finally obtained.

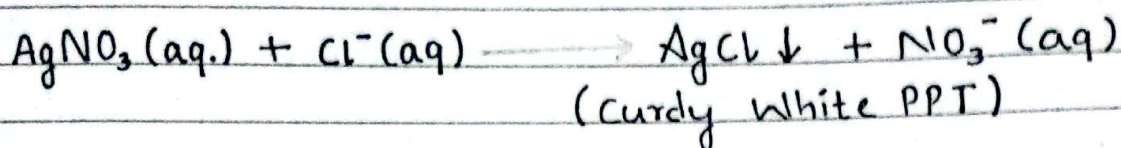
OBSERVATION TABLE

For purity test of mixture of sand and common salt

S.N.	Experiments	Observations	Inferences
1.	The smell of the mixture was taken after heating it.	The mixture had a camphoric smell	The mixture of sand & common salt was impure. i.e. Presence of camphor
After heating the mixture for few minutes			
2.	The mixture was heated for a few minutes and the smell was taken	The mixture had no camphoric smell.	The mixture of sand & common salt was pure. i.e. absence of camphor.

For purity test of sand

S.N.	Experiments	Observations	Inferences
1.	A few drops of AgNO_3 solution were added into the test tube containing 1 ml of unwashed sand solution.	A white precipitate appeared	Sand was not pure, i.e. presence of chloride ion in sand
After washing sand with water several times			
2.	A few drops of AgNO_3 solution were added into the test tube containing 1 ml of washed sand solution.	No white precipitate appeared.	Sand was Pure. i.e. absence of chloride ion in sand.

TEST REACTION**RESULT**

The given mixture was separated into its pure components: camphor was obtained by sublimation, sand by filtration and NaCl by evaporation of filtrate.

CONCLUSION

Hence from above experiment it can be concluded that the mixture containing three component (Volatile, Soluble in water & Insoluble) can be separated by Sublimation, dissolution, filtration followed by evaporation processes.

PRECAUTIONS

1. Personal protective equipments should be worn.
2. During sublimation, mixture should be heated gently.
3. Salt should be dissolved in minimum amt. of water.
4. During filtration, solution shouldn't be poured upto the brim of filter paper present in funnel.
5. Solution should be evaporated just upto dryness, otherwise, the salt may be charred.

VIVA-VOCE

1. What is the chemical formula and systematic name of sand and common salt?
2. What is volatile substance?
3. What is sublimation?
4. What is molecular formula of camphor?
5. Why is camphor volatile?
6. Why is sand nonvolatile?
7. How would you test the presence of NaCl in filtrate in above experiment?
8. What is heterogeneous mixture? Give one example of it.

1) Sand is Silicon dioxide (SiO_2) & Common Salt is sodium chloride (NaCl)

2) Substance that is easily converted into vapour state at ordinary temperature is called Volatile substance. e.g.: Camphor, NH_4Cl

3) Sublimation is the process of conversion of solid directly into vapour by vaporization and condensing that vapor back into solid.

4) Molecular Formula of camphor is $\text{C}_{10}\text{H}_{16}\text{O}$.

5) Camphor molecules are held together by weak Vander Wall's force of attraction which is easy to break at ordinary condition so, camphor is Volatile.

6) Sand is non-volatile because SiO_2 molecules present in sand are held together by strong covalent force and forms giant covalent network of SiO_2 which is very difficult to break at ordinary condition.

7) I would test the presence of NaCl by testing filtrate with AgNO_3 Solution. If filtrate contains dissolved NaCl , then white ppt. of Silver chloride is produced.

8) Particles in mixture are not uniformly distributed is called heterogenous mixture.

e.g.: Sand and common salt, Sand and camphor, and Sand and calcium carbonate etc.

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7/12